

An Introduction to the Stochastic FitzHugh-Nagumo Model

From deterministic excitability to noise-induced spiking

- Project topic: temporal regularity of noise-induced spikes
 - Core model: FitzHugh-Nagumo (FHN)
 - Big question: can noise make spiking more regular?
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Roadmap

1. Why neurons spike
 2. Why use the FitzHugh-Nagumo model
 3. Deterministic FHN dynamics
 4. Adding noise: the stochastic FHN model
 5. New machinery from stochastic calculus
 6. What the simulations suggest
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A little neurobiology

- Neurons communicate with brief voltage pulses called action potentials
 - The membrane potential changes because ions move through channels in the cell membrane
 - After a spike, recovery processes make it harder to spike again immediately
 - So even a simple neuron model needs:
 - a fast voltage variable
 - a slower recovery variable
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Why not just use Hodgkin-Huxley?

- The Hodgkin-Huxley model is biophysically detailed and very successful
 - But it uses several coupled nonlinear equations and many parameters
 - That is great for realism, but harder for qualitative analysis
 - The FitzHugh-Nagumo model keeps the main story:
 - threshold behavior
 - excursion/spike behavior
 - refractory recovery
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The deterministic FitzHugh-Nagumo model

$$\dot{v} = v - \frac{v^3}{3} - w + I$$

$$\dot{w} = \varepsilon (v + a - bw)$$

- v : membrane potential
- w : recovery variable
- I : applied current
- ε : slow-fast time scale parameter

Why this model is useful

- It is two-dimensional, so phase-plane geometry becomes visible
- Nullclines, equilibria, and limit cycles can be studied directly
- It shows several regimes depending on the input current I
- That makes it a good bridge between biology and dynamical systems

Deterministic phase-plane ideas

The nullclines are

$$w = v - \frac{v^3}{3} + I, \quad w = \frac{v+a}{b}$$

- Their intersections give equilibria
- The cubic nullcline creates threshold-like behavior
- The slow recovery variable pulls trajectories back after a spike
- Local stability comes from the Jacobian near equilibrium

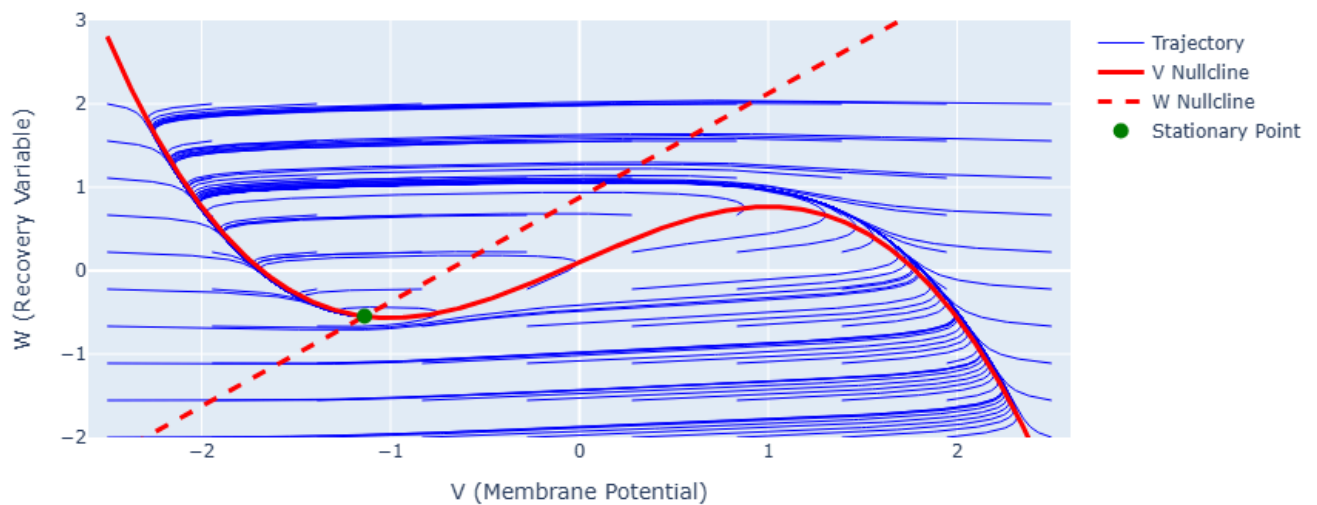
Deterministic regimes

- Small I : stable resting state
- Intermediate I : excitable regime
- Larger I : sustained oscillations / limit cycle spiking

Key idea:

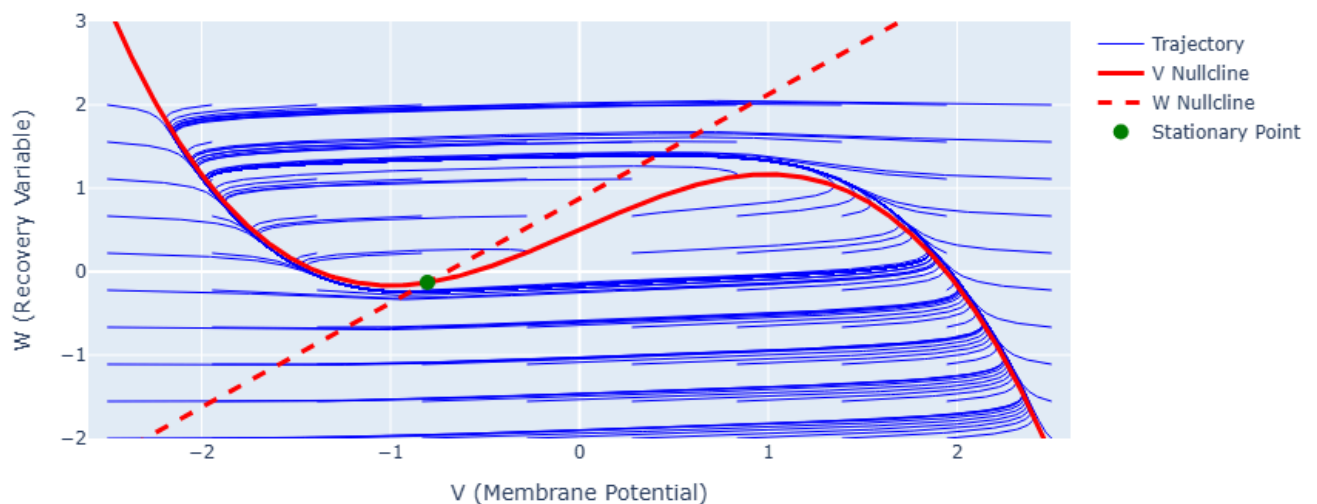
In the excitable regime, the system sits near rest, but a large enough perturbation can trigger a full spike.

Deterministic picture from this project

Phase Plane with Multiple Trajectories ($I_{\text{ext}}=0.1$)

For low input current, trajectories return to a stable equilibrium.

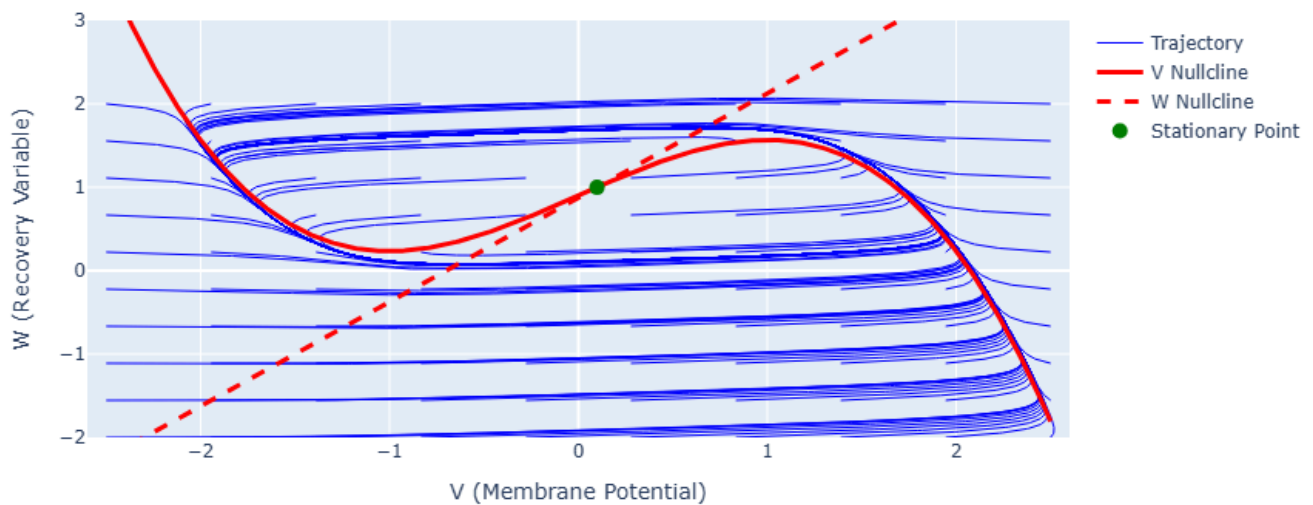
Deterministic picture from this project

Phase Plane with Multiple Trajectories ($I_{\text{ext}}=0.5$)

At intermediate current, the system becomes much more sensitive to perturbation.

Deterministic picture from this project

Phase Plane with Multiple Trajectories ($I_{\text{ext}}=0.9$)



For larger current, trajectories are driven toward repetitive spiking behavior.

Why add noise?

- Real neurons are not perfectly deterministic
- Ion-channel fluctuations, synaptic bombardment, and thermal effects introduce randomness
- In an excitable system, random kicks can push the state across threshold
- So noise can create spikes even when the deterministic system would stay near rest

The stochastic FitzHugh-Nagumo model

$$dV = \left(V - \frac{V^3}{3} - w + I \right) dt + \sigma dW_t$$

$$dw = \epsilon (V + a - bw) dt$$

- W_t : Brownian motion / Wiener process
- σ : noise intensity
- Setting $\sigma = 0$ recovers the deterministic model

What is Brownian motion?

For our purposes, W_t is a random process with:

- $W_0 = 0$
- independent increments
- Gaussian increments: $W_{t+h} - W_t \sim N(0, h)$

- continuous but nowhere differentiable sample paths

Important consequence:

$$\Delta W \sim \sqrt{\Delta t} Z, \quad Z \sim N(0,1)$$

Why stochastic calculus is new here

In a standard nonlinear dynamics course, we usually use:

- ODEs
- phase portraits
- linearization
- bifurcations

For SDEs we need new ideas:

- stochastic integrals
- Brownian scaling
- Ito's Lemma
- numerical methods like Euler-Maruyama

Why ordinary calculus breaks

For a smooth path $x(t)$, we keep only first-order terms in dt .

For Brownian motion, increments are much rougher:

$$(dW_t)^2 \text{ behaves like } dt$$

So second-order terms involving noise do not vanish.

That is the source of Ito corrections.

Ito's Lemma: statement

Suppose

$$dX_t = \mu(X_t, t)dt + \sigma(X_t, t)dW_t$$

and $f(x, t)$ is smooth. Then

$$df(X_t, t) = \left(f_t + \mu f_x + \frac{1}{2} \sigma^2 f_{xx} \right) dt + \sigma f_x dW_t$$

Compared to the chain rule, there is an extra term:

$$\frac{1}{2} \sigma^2 f_{xx} dt$$

Ito's Lemma: where the extra term comes from

Start with a second-order Taylor expansion:

$$df \approx f_t dt + f_x dX_t + \frac{1}{2} f_{xx} (dX_t)^2$$

Now insert

$$dX_t = \mu dt + \sigma dW_t$$

Then

$$(dX_t)^2 = \mu^2 (dt)^2 + 2\mu\sigma dt dW_t + \sigma^2 (dW_t)^2$$

Use the Ito rules:

$$(dt)^2 = 0, \quad dt dW_t = 0, \quad (dW_t)^2 = dt$$

So only $\sigma^2 dt$ survives.

Why Ito's Lemma matters here

- It replaces the chain rule for stochastic systems
 - It is the basis for analyzing transformed stochastic processes
 - It explains why noisy dynamics are not just ODEs with a random forcing term
 - It is one of the first genuinely new technical tools beyond deterministic dynamical systems
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Numerical method: Euler-Maruyama

For small time step Δt ,

$$v_{n+1} = v_n + \left(v_n - \frac{v_n^3}{3} - w_n + I \right) \Delta t + \sigma \sqrt{\Delta t} Z_n$$

$$w_{n+1} = w_n + \epsilon (v_n + a - b w_n) \Delta t$$

with $Z_n \sim N(0,1)$.

- This is the stochastic analogue of forward Euler
 - The $\sqrt{\Delta t}$ factor comes from Brownian scaling
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What quantity are we trying to optimize?

We measure spike-time regularity using the coefficient of variation of inter-spike intervals:

$$CV = \frac{\sigma_{ISI}}{\mu_{ISI}}$$

- Small CV = more regular spike timing
- Large CV = more irregular spike timing
- If an intermediate noise level minimizes CV, that suggests coherence resonance

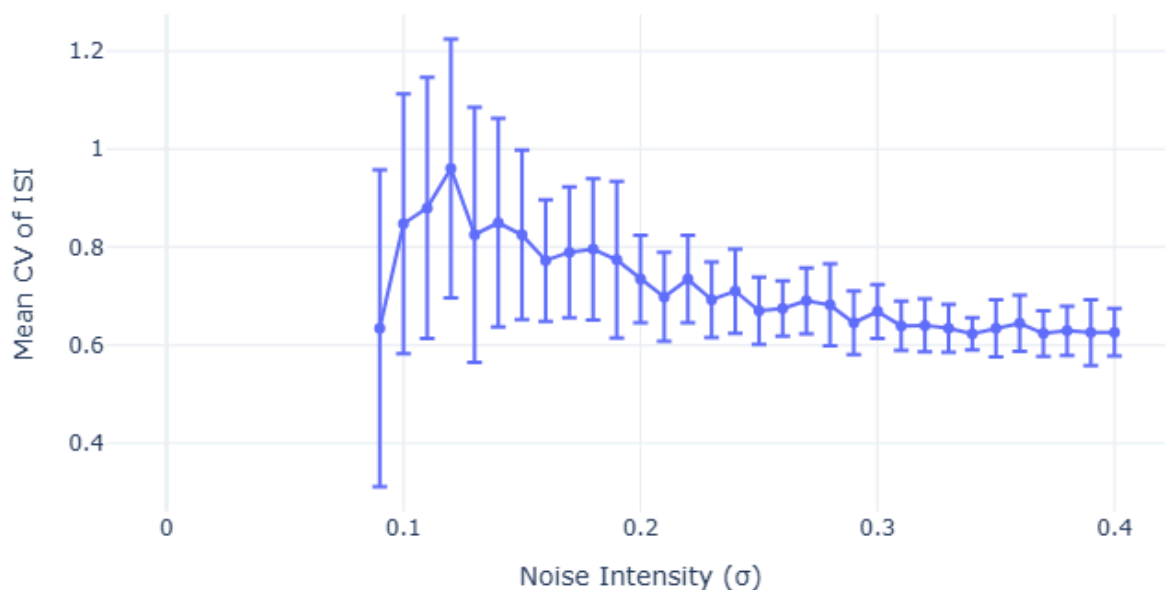
Coherence resonance

- Too little noise: spikes are rare and irregular
- Too much noise: the system is pushed around too strongly and timing becomes irregular again
- Intermediate noise: threshold crossings can become surprisingly regular

So noise is not always destructive; sometimes it organizes the dynamics.

Simulation result: CV versus noise

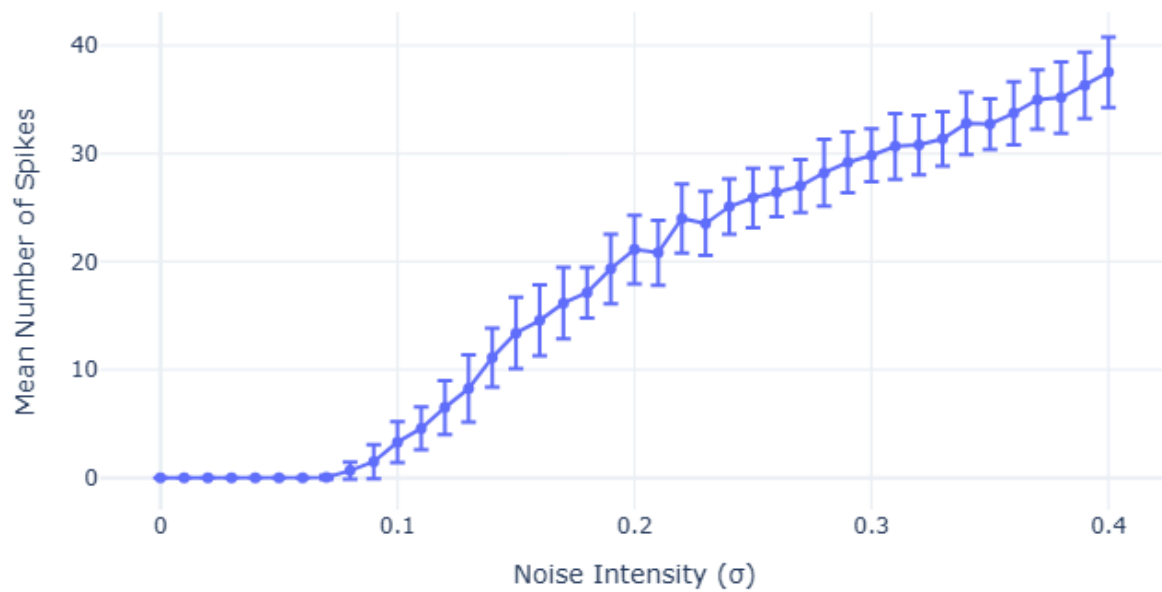
Mean Coefficient of Variation of ISI vs Noise Intensity



- The curve suggests a dip at intermediate noise intensity
 - That is the signature we expect for coherence resonance
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Simulation result: spike count versus noise

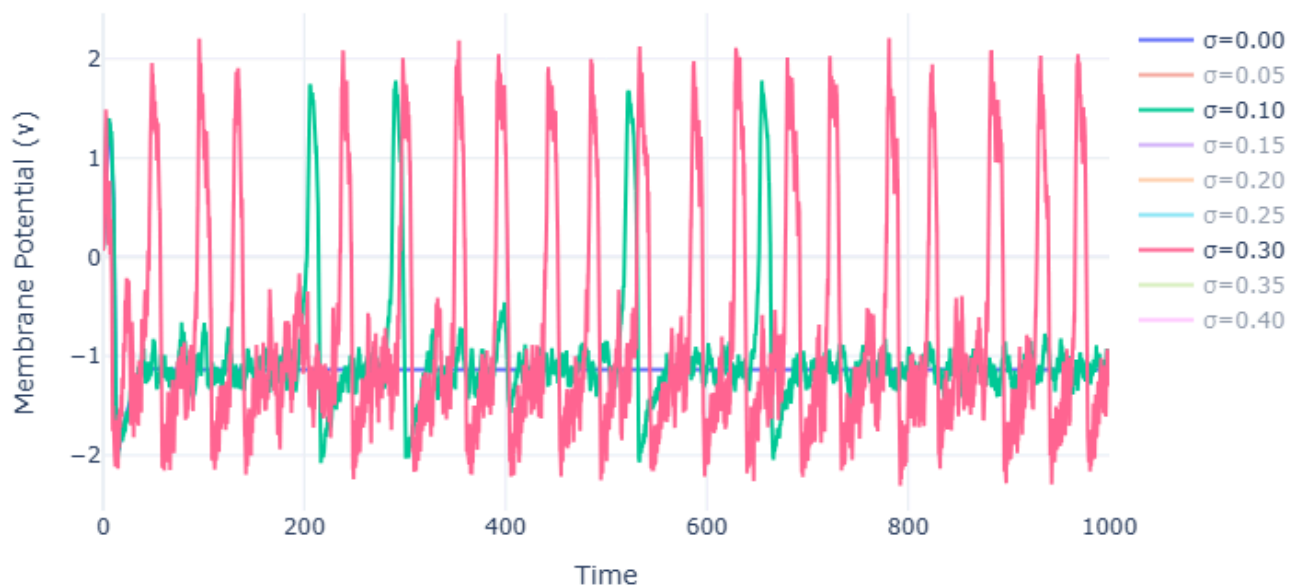
Mean Spike Count vs Noise Intensity



- As noise intensity grows, spike counts increase
- More noise makes threshold crossings easier
- But more spikes does not automatically mean more regular spikes

Simulation result: sample trajectories

Sample Membrane Potential Trajectories for Selected Noise Intensities



- Low noise: small fluctuations near rest
- Moderate noise: occasional and more structured spikes
- High noise: frequent but more strongly perturbed activity

Deterministic vs stochastic viewpoint

Deterministic model:

- asks about equilibria, limit cycles, and basins

Stochastic model:

- asks about distributions, random trajectories, spike statistics, and mean behavior across trials

Same system, but a different language of analysis.

Main takeaways

- FHN is a reduced neuron model with clear geometric structure
- The deterministic system explains excitability and repetitive spiking
- Noise can induce spikes even in a resting deterministic regime
- Stochastic calculus is needed because Brownian motion is too rough for ordinary calculus
- Ito's Lemma and Euler-Maruyama are the main new tools at this level
- The project data is consistent with coherence resonance

Possible concluding question

How does randomness interact with nonlinear threshold dynamics so that an intermediate amount of noise can make a neuron fire more regularly rather than less regularly?

Thank you

Possible next directions:

- compare Ito and Stratonovich interpretations
- discuss first-passage times and spike thresholds
- connect to coherence resonance in other excitable systems